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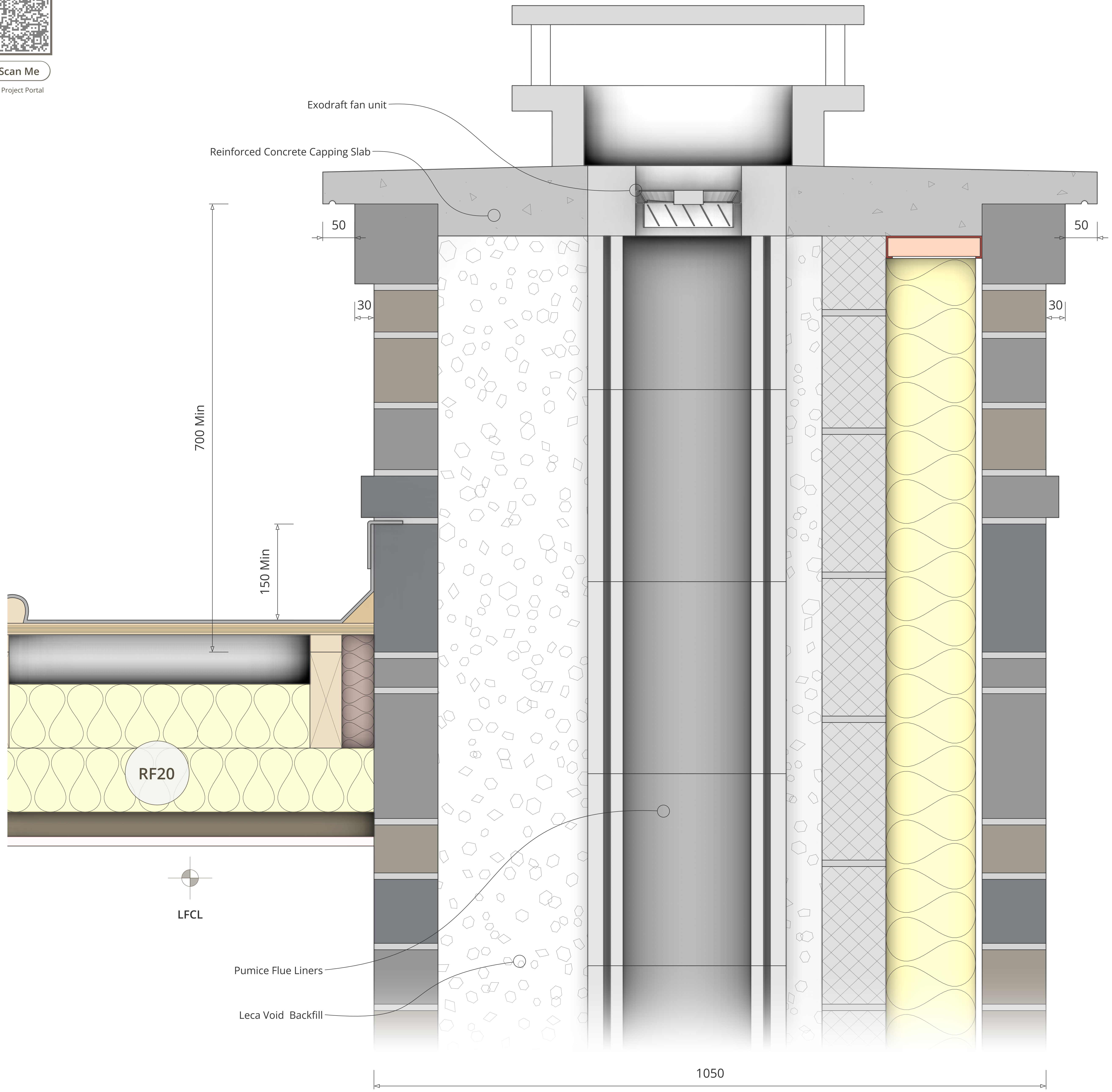

Element	Specifications
Reinforced Concrete Capping Slab	Construct a Cast-in-Situ Reinforced Concrete Capping Slab to terminate the chimney stack. Dimensions: Minimum 100mm thick x Full width of chimney breast plus overhang. Reinforcement: Steel mesh (A142 or similar) set centrally with 40mm cover. Overhang: Form a minimum 50mm projection beyond the outer stone face on all sides to shed water; include a formed 6mm 'drip throat' or groove on the underside of this overhang (minimum 25mm from the stone face) to prevent water running back down the masonry. Surface Finish: Trowel to a smooth, dense finish with a very slight fall away from the centre to discourage standing water. This slab provides the structural base for the Exodraft fan unit. [Compliance: BS EN 1992-1-1]
Chimney Void Backfill	Fill the annular void between the Isokern Outer Casing and the external Stone/Block skin with a lightweight insulating mix. Material: Leca (Lightweight Expanded Clay Aggregate) mixed with a weak cement slurry (Ratio approx 20:1 Leca to Cement). Do NOT use loose vermiculite or standard heavy concrete. Installation: Pour in 'lifts' (stages) of maximum 1.5m height per day to prevent the hydrostatic pressure of the wet mix from blowing out the blockwork or crushing the inner liner. Purpose: This 'no-fines' mix stabilizes the flue column, provides thermal insulation to keep gases hot, and allows moisture to drain away rather than soaking into the structure.
Flue Offset & Support Lintel	Construct the required flue deviation (Front-to-Centre) using specific Isokern DM 30° Offset Blocks (or manufacturer specific 15°/30° components to achieve the trajectory). Support: Install a 150mm (h) x 100mm (w) Pre-stressed Concrete Lintel immediately below the first offset block to carry the load of the deviated stack above. Ensure this lintel has a minimum 150mm bearing onto the structural blockwork at either side. Jointing: Bond all offset blocks using Isokern Lip Glue, checking internal smoothness to prevent turbulence. Warning: Do not offset at an angle greater than 45° from the vertical (Building Regs Part J limit).
Electrical Conduit	Install a 25mm flexible PVCu or Steel Conduit within the chimney void *before* pouring the LECA backfill. Routing: Run the conduit from the Firechest Gather/Switch point, strapped securely to the outside of the Isokern Casing, terminating flush with the top surface of the Concrete Capping Slab. Warning: Failure to install this conduit now will make wiring the Exodraft fan impossible later. Ensure a draw-rope is left inside the conduit for the electrician.
Fan Mounting Interface	Finish the Concrete Capping Slab with a perfectly flat, level mounting zone around the flue exit to accept the Exodraft Fan Flange or Stainless Steel Shroud. Sealing: Create a movement joint between the top of the pumice liner and the concrete slab using a 10mm foam expansion strip or ceramic fibre rope; do not cast concrete directly against the pumice liner as thermal expansion will crack the slab. The final weather seal will be achieved by the high-temperature silicone sealant and the fan's base flange gasket.
Installation	General: Ensure the stone outer leaf is fully completed and cured before casting the heavy concrete cap to prevent compression of green mortar. Weathering: If the 'Stainless Steel Box' shroud is a separate aesthetic cover, ensure it is mechanically fixed to the slab using stainless steel resin anchors and that the junction is sealed against wind-driven rain.
Important	The 'Wet' Mix: When mixing the LECA concrete, use the minimum amount of water possible (semi-dry consistency). If the mix is too wet, the cement slurry will run out of the Leca and block the weep holes or seep into the liner joints. Structural Safety: Ensure the temporary formwork for the concrete capping slab is robustly propped during the pour and cure period (min 7 days).
Design Rationale	The design utilizes a reinforced concrete capping slab rather than a traditional flaunched mortar top. This provides a robust, level structural platform essential for the mechanical fan unit and prevents the common failure of cracked mortar allowing water ingress. The use of LECA concrete backfill ensures the flue liners are insulated (improving fan efficiency) and structurally rigid, while the dedicated lintel transfers the offset loads safely to the masonry shell.
Standards	BS EN 1858 (Chimney Blocks), BS EN 1992 (Concrete Structures), Approved Document J (Combustion Appliances), BS 7671 (Electrical Wiring - Conduit).

Building Control Approval Submission

This drawing is submitted for Building Control approval. Construction must not commence until the Inspector sanctions the proposed structural details, thermal performance, and moisture integrity. The Principal Contractor is fully responsible for coordinating all statutory inspections to verify strict compliance with the Building Regulations 2010.

CM40 - Chimney Capping Details

00 100 200 300 400 500

Printed Scale Factor | 1mm : 5mm   Scale Valid When Printed At : Iso A2